

VICTOR NGO

@ CONTACT

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📍 Murrieta, CA

🌐 <https://vngosb.github.io>

🎓 EDUCATION

Masters of Science

Bioinformatics

Johns Hopkins University

Bachelors of Science

Biology

**University of California,
Riverside**

★ SKILLS

- **Genomics & NGS:** RNA-seq analysis, germline variant calling, SNP/variant analysis, transcriptomics, genomic data interpretation, differential expression analysis
- **Tools & Platforms:** Nextflow, DESeq2, Salmon, BWA, GATK, HISAT2, BEDTools, SAMtools, IGV, Galaxy, Docker
- **Programming & Computing:** Python, R, SQL, Bash, Linux/UNIX
- **Clinical Diagnostics & Assay Analytics:** Assay performance analysis, validation & verification, NGS QC, CAP/CLIA-regulated environment, root cause analysis, statistical trend analysis
- **Data Analysis & Visualization:** pandas, NumPy, ggplot2, statistical analysis, PCA, heatmaps, volcano plots

📄 PROFESSIONAL SUMMARY

Bioinformatics professional with experience developing reproducible NGS workflows for RNA-seq and genomic data processing using Nextflow, Python, R, and Linux-based tools. Background in molecular diagnostics and statistical assay analysis within CAP/CLIA-regulated environments, supporting scalable bioinformatics and diagnostic workflows.

👤 WORK EXPERIENCE

QC Specialist II, Data Analysis

GENMARK DIAGNOSTICS (Roche)

📅 2022 - current

📍 CARLSBAD, CA

- Analyzed large-scale multiplex molecular diagnostic datasets using statistical trend analysis to identify assay variability, out-of-specification (OOS) events, and performance trends across manufacturing lots and reagent conditions
- Investigated assay performance issues by correlating diagnostic outputs with manufacturing and process variables to support root cause analysis, corrective actions, and process improvements
- Collaborated cross-functionally with QA, R&D, Software, Manufacturing, and Data Analytics teams to troubleshoot and resolve system-level issues impacting diagnostic assay performance and reliability
- Performed validation and verification activities for diagnostic assays and reagents within a CAP/CLIA-regulated environment, supporting data integrity and assay reproducibility

❤️ PROJECTS

Germline Variant Calling Pipeline (In Progress)

- Developing reproducible NGS workflow using BWA-MEM, SAMtools, and GATK for alignment, BAM processing, and germline variant calling from whole exome sequencing data using public Illumina datasets

Nextflow RNA-Seq Pipeline (NASH vs Control)

- Developed a reproducible end-to-end Nextflow RNA-seq pipeline integrating quality control, alignment, transcript quantification, and differential expression analysis of public NGS datasets, identifying differentially expressed exosomal miRNAs associated with NASH

RNA-Seq Differential Expression Pipeline

- Developed an RNA-seq differential expression workflow using Salmon and DESeq2 to analyze tumor versus normal samples, generating PCA, heatmaps, and volcano plots to visualize transcriptomic differences and expression patterns.